

Performance Testing

Date	4 July 2026
Team ID	
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Python unittest, Flask Test Client
Type of Testing	Performance Testing, Load Testing, Integration Testing
Target Module / API	Flask Web Application (/predict, /api/predict)
Test Environment	Local Machine (Windows, Python 3.x)
Test Date	4 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Single Prediction Request	1	5	Response within target time
2	Batch Prediction (100 Requests)	1	30	Stable performance with no failures
3	Invalid Input Handling	1	5	Returns proper validation error (HTTP 400)

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	35.04 ms	Pass	Excellent response time
2	Response Time (Max)	< 5 seconds	< 1 sec	Pass	Within acceptable limit
3	Throughput (Req/sec)		≈33.7 req/sec	Pass	Based on 100 requests in 2.97 seconds
4	Error Rate	< 1%	0%	Pass	All tests passed successfully
5	CPU Utilization	< 80%	Within acceptable limit	Pass	No abnormal CPU usage observed
6	Memory Utilization	< 80%	Within acceptable limit	Pass	Stable memory usage

Step 4: Observations & Analysis

Key Findings:

The application successfully passed all 16 test cases. It achieved an average prediction latency of approximately 35 ms, handled invalid inputs correctly, and maintained stable performance during batch prediction.

Bottlenecks Identified:

No significant performance bottlenecks were observed during testing.

Optimization Steps Taken:

Optimized preprocessing logic, reused the trained model in memory, validated inputs before prediction, and eliminated redundant computations.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

```
40 | else:
    Problems    Output    Debug Console    Terminal    Ports
Outliers in 'annual_income' capped at IQR bounds: [0.00, 0.00]
Outliers in 'years_employed' capped at IQR bounds: [0.00, 0.00]
Outliers in 'debt_to_income' capped at IQR bounds: [0.00, 0.00]
Removed 0 duplicate records.
Outliers in 'annual_income' capped at IQR bounds: [0.00, 0.00]
Outliers in 'years_employed' capped at IQR bounds: [0.00, 0.00]
Outliers in 'debt_to_income' capped at IQR bounds: [0.00, 0.00]
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Ran 16 tests in 3.761s

OK
PS C:\Users\anand kumar\OneDrive\Desktop\Credictcardapprovalpredicationsystem>
ApexTrust Auditor (2 days ago) OVR Ln 6, Col 1 Spaces: 4 UTF-8 LF {} Python 3.13.5 64-bit
```